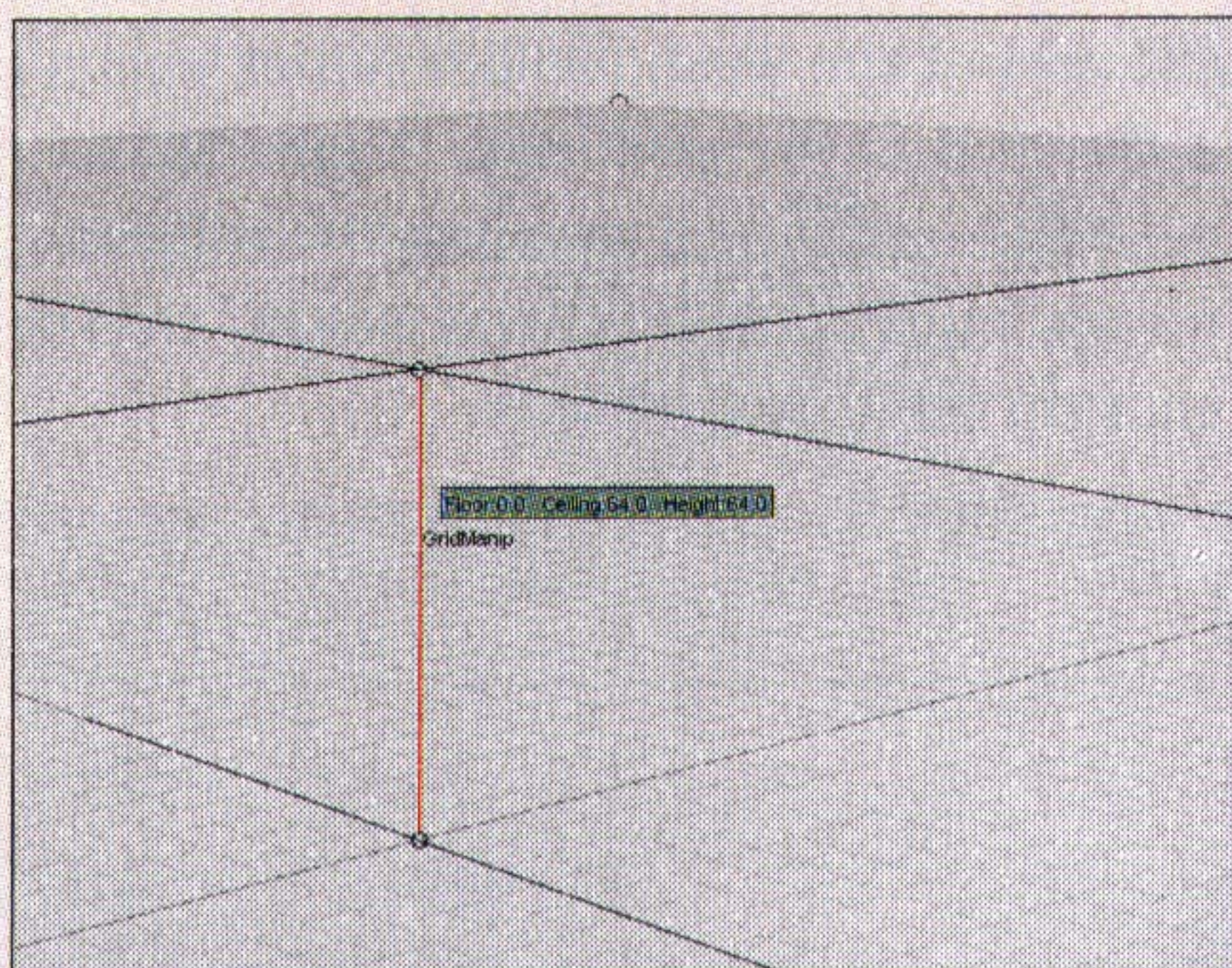




To change the height of the grid map, click the Brush Manipulate button and hover your mouse over the Height Grid Manipulator—look for a long vertical line with circles at its ends. This adjusts the position of the grid object in space—you can move it up, down or side ways by either using the circles or the line itself.



The Height Grid Manipulator adjusts the grid object in space



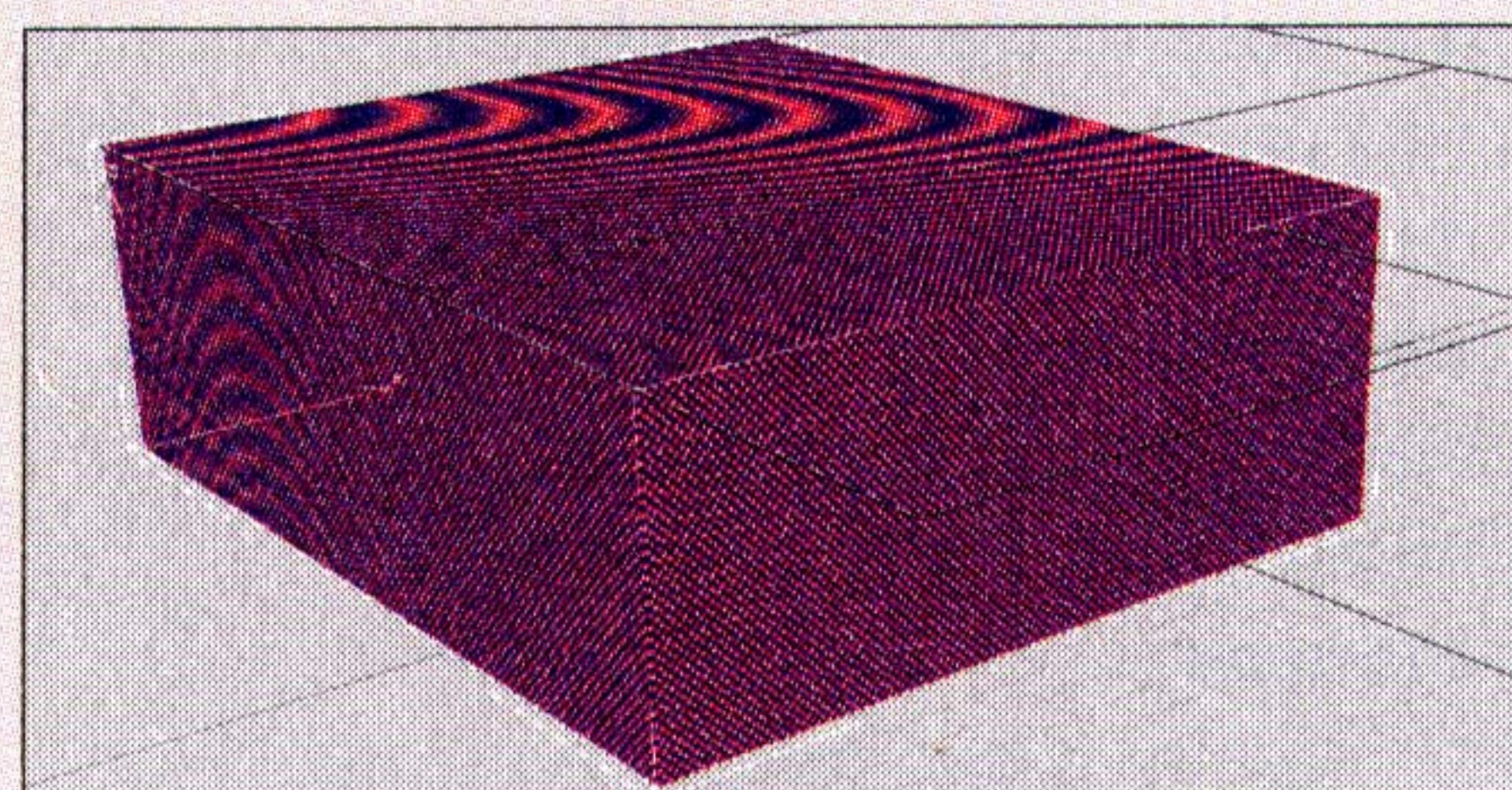
To adjust the height of the Height Grid, click and hold on the upper circle—a tool tip will indicate the height dimension above this circle. Click-hold on the circle and drag upward until the dimension reads 'Floor: 0.0 Ceiling: 512 Height: 512'.



Observe the Grid Settings buttons on the lower right-hand part of the screen. Click on the button labelled 64—this is the one we will use to create the level. Next, click on the Create Brush button—this will put you in brush create mode.



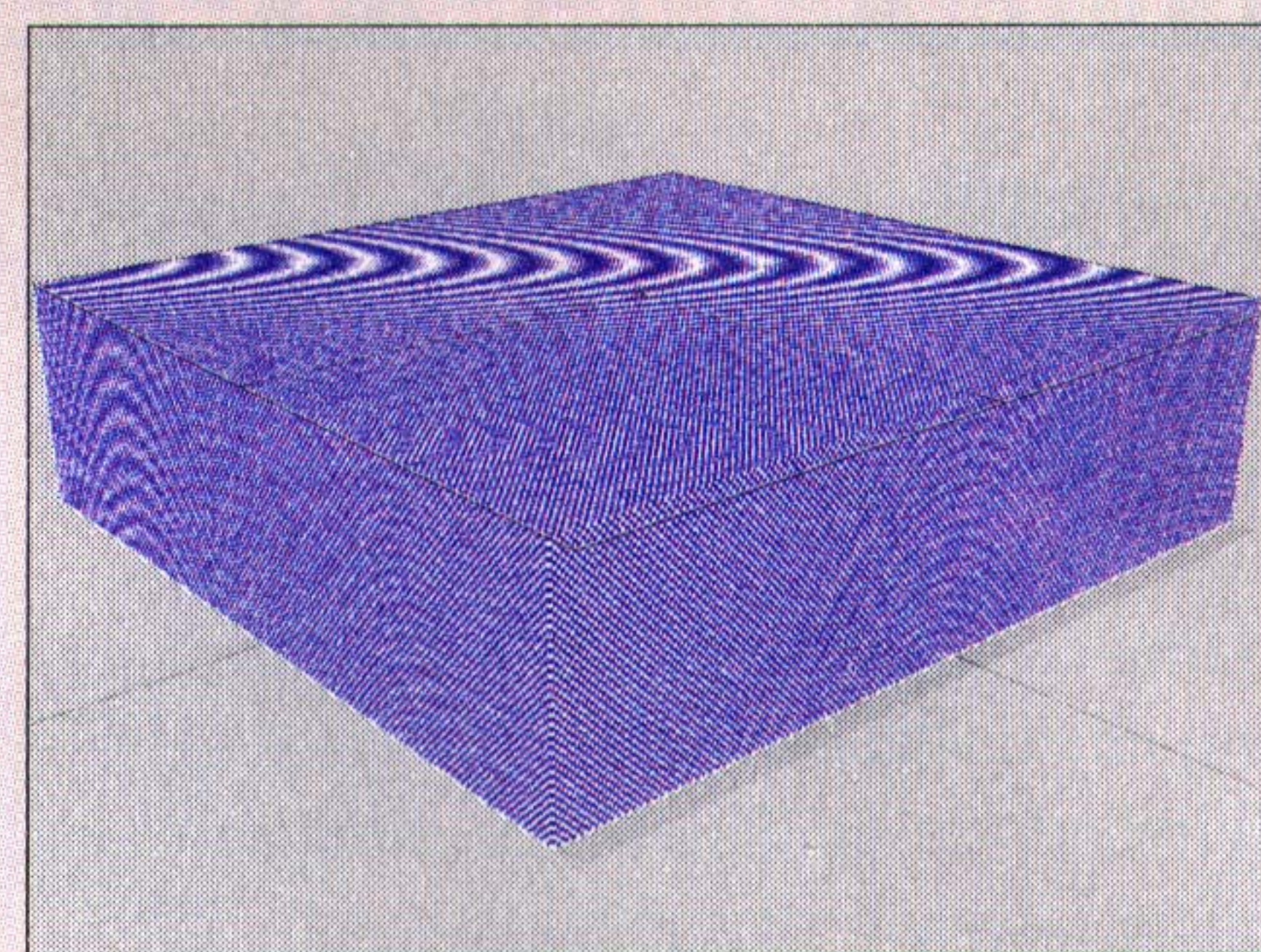
There should be a light blue cross hair on the grid object. If not, then click the Snap button to active grid snaps. Zoom out to view the entire grid. Aim your cursor in the far left corner of the grid, click-hold, drag, and then release the button at the right corner on the opposite side. This will create a big box (a brush) with a checkerboard pattern.



Create Brush button to create a brush object



Choose the number 4 or 2 Grid Spacing button and then make the brush object hollow by clicking on the CSG Hollow button. This will create an empty level. Click on the QuakeCam button and then inside the empty level—move within the box using the



The brush created should now be hollowed. This is your primordial level

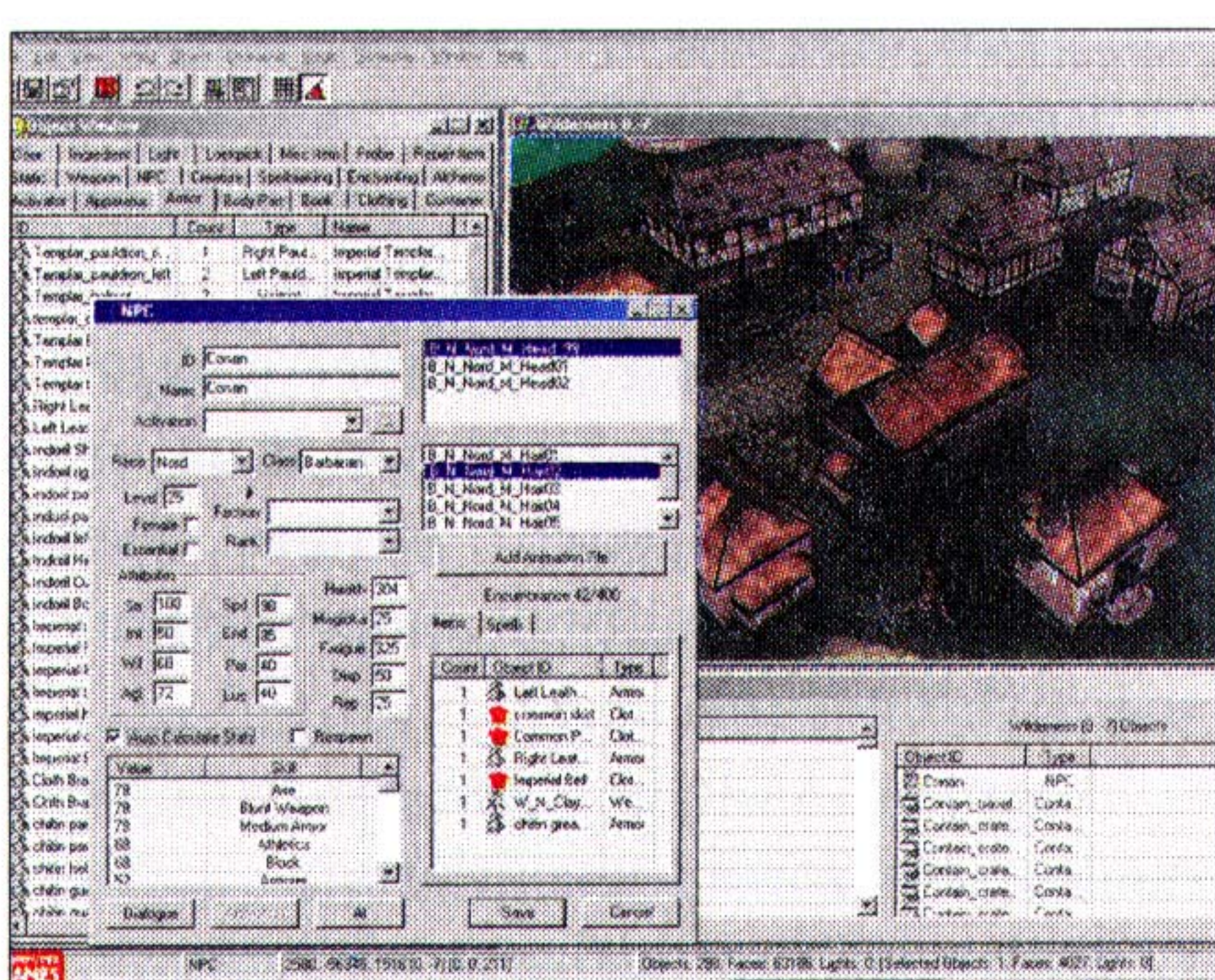
[W] + [A] + [S] + [D] + mouse combination.



To apply textures to the walls, click on the Shader Nav Button. This opens the Quake Shader Navigator tool. From the Shader pull-down menu, choose the base_wall shader category.



Click on the Select tool in the ToolBar. This will enable you to select walls and floors to apply textures—notice that a selected surface turns red. Click on one of the vertical walls and for multiple selections, press down the control button while clicking. You should be able to click on any of the thumbnails in the Shader Nav to apply a texture to your selection.



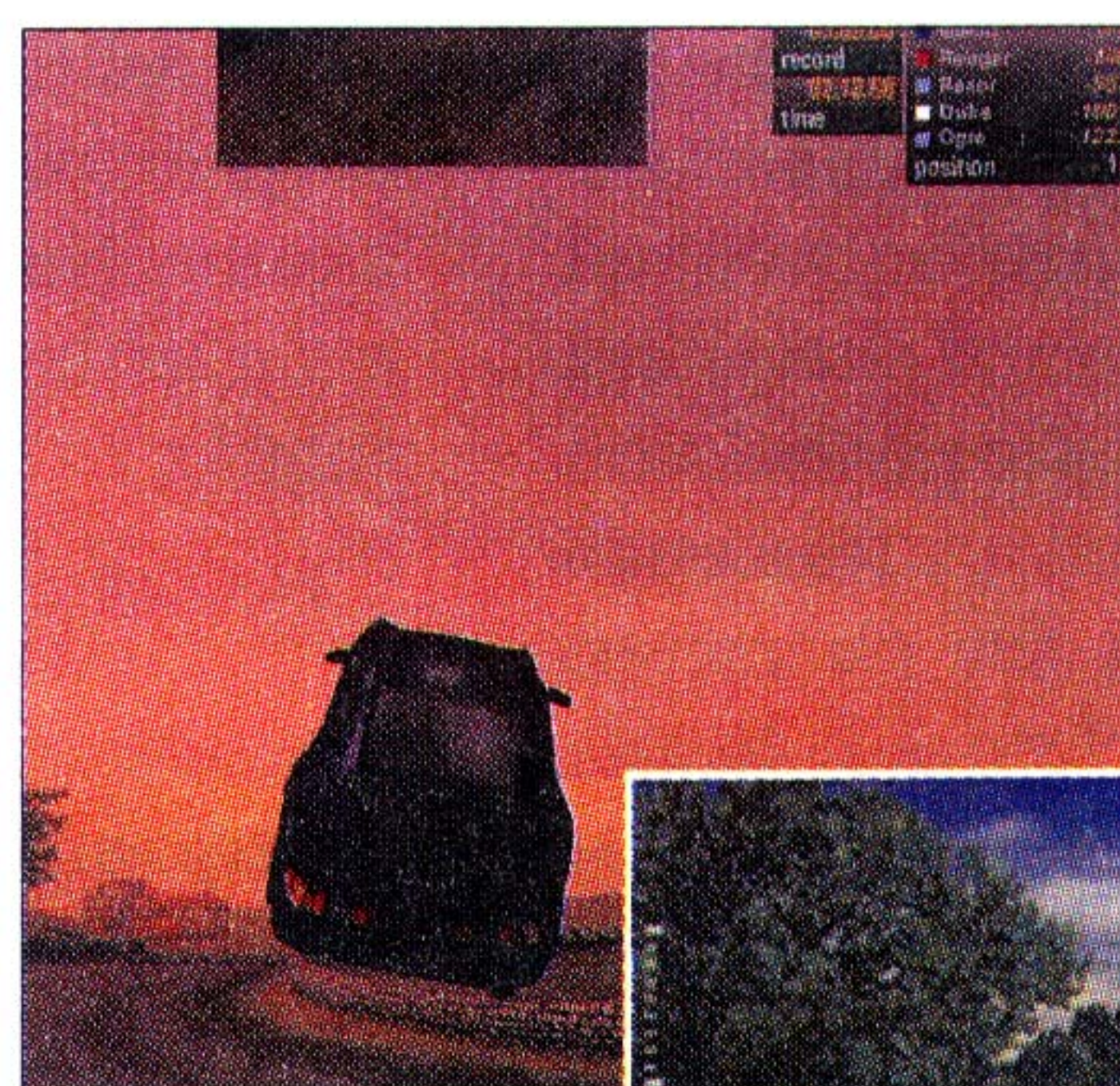
Most games come bundled with their own level designing tools. This is Bethesda Softwork's TES Construction Set

possess none of these skills. There are enough resources on the Internet that will help you get started on level designing (see box, 'Get your kicks here') and our small workshop on creating a level for *Quake III Arena* using the gmax tool ('This used to be my Playground') will help you get off the ground. Formally trained or not, practise

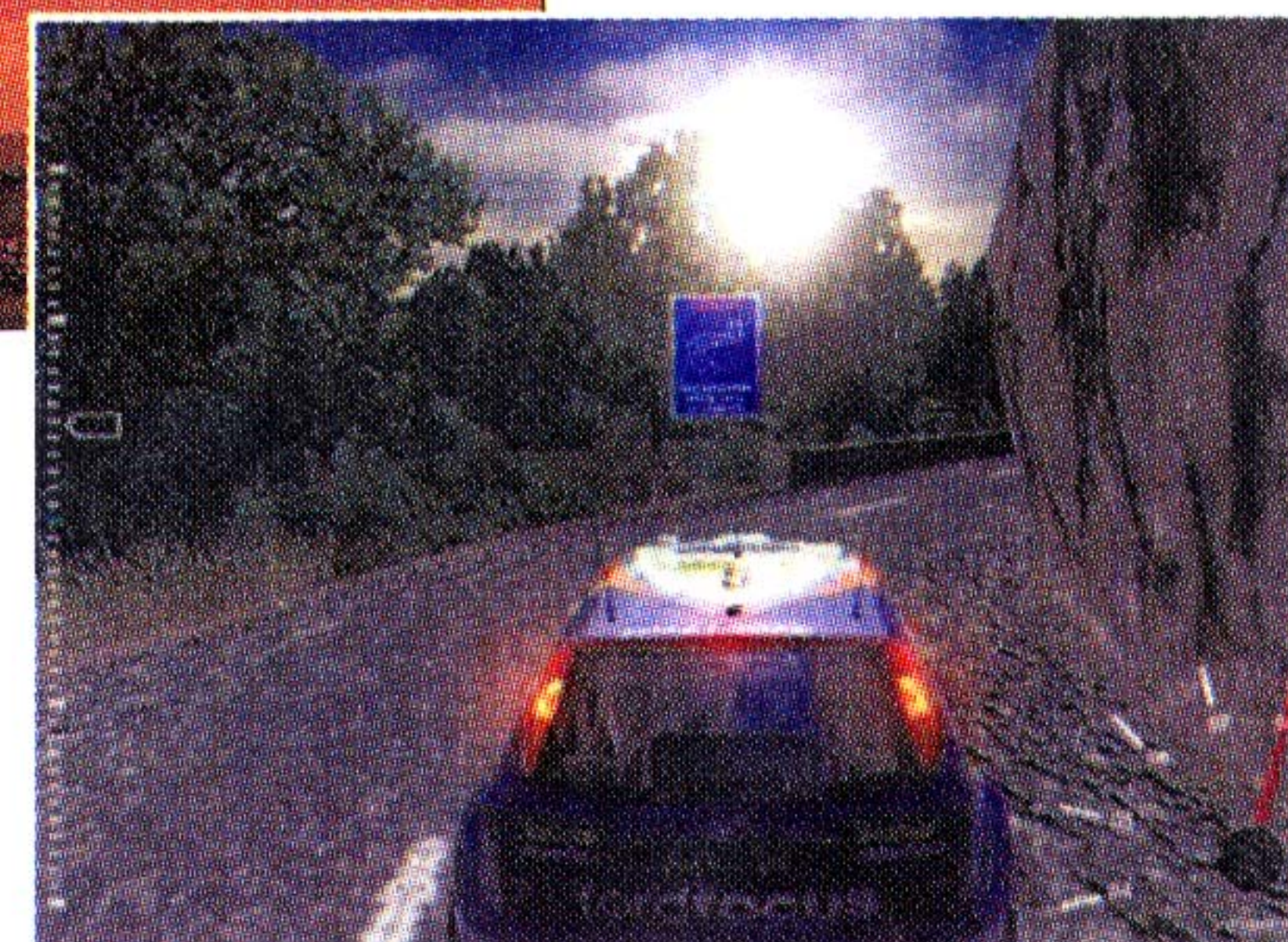
makes a great level designer. So pick up a game like *Unreal Tournament*, open its levels, study them and attempt to learn from the masters.

A level head

Making a game is no longer a solo venture and team play implies good communication. A level designer should know what is required from his level, design-wise, even as the polygon-budget is foremost in his mind. The task becomes a little more complex if the mapper is also going to create his own textures—you would have to ensure that they gel visually with the rest of the game. Often, a level creator will first prepare a concept sketch, clear it with the rest of the team and only then touch mouse. If you think that you have what it takes to be a level designer, what follows are some pointers that will help you.



'While taking to the air is always fun, nothing keeps a player guessing like a blind corner



■ A racing game should supply a speed-induced rush; this need not mean long, fast straights. There is nothing more exhilarating than a tire burn at 300 kph as you hit the handbrake to execute a 180-degree turn into a tunnel. Intersperse